

$\therefore$ By Defn. 2, the sets are not equal.

c) $\emptyset, \{\emptyset\}$
Soln: By Example 12, we know $|\emptyset|=0$. $|\{\emptyset\}|=1$. $\therefore$ The second set has 1 element. By Defn 2, the sets are not equal.

8) Suppose that $A=\{2,4,6\}$, $B=\{2,6\}$, $C=\{4,6\}$ and $D=\{4,6,8\}$. Determine which of these sets are subsets of which other of these sets.

Solution: 2 and 6 are the only elements of B. $2 \in A$ and $6 \in A$.
By Defn 3, $B \subseteq A$. $4 \in A$ but $4 \notin B$. $\therefore$ By Defn 3, $A \not\subseteq B$.

4 and 6 are the only elements of C. $\therefore$ By Defn 3, $C \subseteq A$.

$2 \in A$ and $2 \notin C$. $\therefore$ By Defn 3, $A \not\subseteq C$.

$2 \notin D$. $\therefore$ By Defn 3, $A \not\subseteq D$. $8 \in D$ but $8 \notin A$. $\therefore$ By Defn 3, $D \not\subseteq A$.

It's easy to see why $B \not\subseteq C$ and $C \not\subseteq B$.
We have seen that $2 \notin D$. By Defn 3, $B \not\subseteq D$. $4 \notin B$ but $4 \in D$. $\therefore$ By Defn 3, $D \not\subseteq B$.

$4 \in D$ and $6 \in D$. $\therefore$ By Defn 3, $C \subseteq D$. $8 \in D$ but $8 \notin C$. $\therefore$ By Defn 3, $C \not\subseteq D$.

$\therefore$ Final Answer: $B \subseteq A, C \subseteq A, C \subseteq D$

9) For each of the following sets, determine whether 2 is an element of that set.

a) $\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$
 $2 \in \mathbb{I}$ and $2 > 1$. $\therefore 2$ is in the set

b) $\{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$
 2 is not the square of any integer. $\therefore$ No

c) $\{2, \{2\}\}$. Yes

d) $\{\{2\}, \{\{2\}\}\}$ $\{2\}$ and 2 are not the same. $\{\{2\}\}$ and 2 are not the same. $\therefore$ No

e) $\{\{2\}, \{2, \{2\}\}\}$ $\{2\}$ and 2 are not the same. $\{2, \{2\}\}$ and 2 are not the same. $\therefore$ No